

Each of the subitems below will be rated from 1-5:

- 1 = very positive aspect of this paper
- 2 = positive aspect of this paper
- 3 = neutral or irrelevant criterion for this paper
- 4 = negative aspect of this paper
- 5 = very negative aspect of this paper

YOU MUST ALSO PROVIDE COMMENTS TO THE AUTHORS IN PROSE. IT IS NOT ACCEPTABLE TO MERELY FILL OUT NUMBERS, AND RETURN THE REVIEW!

Paper Title: EchoStature: Spatial Demographic Conditioning for Pediatric Ejection Fraction Regression, with Subgroup-stratified Evaluation

First Author Last Name: Narayan

Appropriateness for PSB
Is relevant to conference themes :1
Is relevant to session themes :1
Meets length restriction (12 pages):1

Comment:
This paper is highly relevant to PSB because it combines biomedical AI and important evaluation/fairness problem in pediatric imaging. It also fits well with the session themes, mainly focusing computational methods using multimodal data and demographic subgroup evaluation.

Presentation
Is clearly written :2
Appropriate title and abstract :2
Adequate figures and tables :2
Clear problem statement (inputs and outputs) :1
Clear descriptions of algorithms :2
Provides appropriate detail :2
Describes scope and limitations of the approach :1
Describes implementation status :2
Adequate references :2

Comment:
This paper has a very strong overall presentation. Starting with a clear goal and objective, input and output (ECG clip and demographics to predict EF regression), clear and detailed figures and captions, and algorithm details. My main request would be for more explicit clarification of the train/validation split and whether it was per patient or per clip, since this is important for interpreting the results and avoiding data leakage (so it needs to be per-patient).
Significance
Represents a biologically important advance :3
Addresses a biologically important and realistic problem :2
Represents a computationally important advance :2
Addresses a computationally important problem :2
Is likely to scale up to realistic problems :3
Provides new evidence for an existing technique :1

Comment:
The most significant contribution would be the subgroup-stratification findings, where subgroup variation is much larger than model-to-model difference. However, this paper is still in an early stage so real-world implementation/scalability is not established yet.

Originality
Represents a novel approach or combination of approaches :2
Has not been published before :1
Points out differences from related research :1
Reformulates a problem in an important way :1

Comment:
This paper identifies gaps in the current literature very early on and their objectives are based on the gaps. Very interesting original idea in this pediatric EF context that treats demographic context as spatial structure rather than a simple scalar metadata input and use them to generate synthetic data.

Technical Content
Evaluates the effectiveness of implementation or techniques :1
The approach is supported with sound arguments :1
The approach is supported with theoretical analyses :2
The approach is supported with experimental results :1
The approach is technically sound :2

Additional Comments for the Authors:

Technically strong work overall. My only minor concern is the evaluation split: I would like explicit confirmation that the validation was per-patient and internal versus external. If the split was at clip level only, the results could be less robust than they appear. Also please verify the github repo link.

Overall recommendation:

ACCEPT MARGINAL REJECT

Your confidence in recommendation:

STRONG MEDIUM WEAK

I used secure LLM to refine the language of my review and I confirm that I take full responsibility for this review.

Reviewer LLM Attestation

If secure LLMs were used to create this review, please provide a brief statement describing how you used LLMs and confirm that you take full responsibility for this review. The use of secure LLMs is not forbidden. The PSB organizers would like to track the use of secure LLMs over time. Please note that your attestation will be provided to the authors, so please do not provide any identifying information in your attestation in order to maintain your anonymity.

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